

Bounds on the Dynamic Causal Effects of Monetary Policy

An Analysis Using the Federal Reserve's Textual Repository

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Abstract

Semiparametric estimates of the effects of monetary policy treat each FOMC meeting as a unit that selects into a policy choice, and require every choice to be possible at every point of the covariate space (overlap). Using the policy alternatives that Federal Reserve staff lay out in the Bluebook before each meeting, I show that overlap fails in a specific, observable way: over 133 meetings from 1989 to 2006 the Committee never took an action that was absent from its menu. I build a propensity-score model in which the menu imposes these structural zeros, derive the sharp identified set for the average dynamic effect of a rate increase or decrease when some menus exclude the action or the status quo, and propose a one-parameter relaxation that trades the worst case for an assumption on effect heterogeneity across menus. The menu constraint alone improves the fit of the policy model as much as adding market expectations. On the empirical side, worst-case bounds are uninformative, and the effects that are point-identified on the meetings with genuine overlap have the wrong sign for industrial production and unemployment even after conditioning on the Greenbook forecasts, although the forecasts do remove the price puzzle. Handling overlap does not rescue selection-on-observables in this setting: the binding problem is unconfoundedness.

Note on this edition. This revision replaces the version submitted in May 2020. The conceptual idea is unchanged; the estimation was revised and the code rebuilt from the raw data. The main changes are the propensity score, the calibration of the outcome bounds, the treatment of the point estimates, and the addition of inference. Several conclusions change as a result. Appendix C compares the two versions. All numbers here are produced by the pipeline in the accompanying repository.

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1 Introduction

Measuring the effect of monetary policy on the economy is one of the central problems of empirical macroeconomics. The difficulty is familiar. Policy responds to the state of the economy, which in turn responds to policy, and the econometrician sees a few hundred monthly observations of this feedback loop. The dominant response has been to impose structure, through structural vector autoregressions (Sims, 1980) or local projections (Jordà, 2005), and with it linearity in the effect of policy.

A second approach, developed by Angrist and Kuersteiner (2011) and Angrist et al. (2018), borrows from microeconomic policy evaluation. Each FOMC meeting is a unit that selects into a policy choice (ease, hold, or tighten), and the effect of that choice is recovered by inverse probability weighting (IPW) on a propensity score for the Committee’s decision. The approach is semiparametric and allows effects to be nonlinear and asymmetric. Like all selection-on-observables designs, however, it rests on two assumptions. *Unconfoundedness* requires that, given the observed covariates, the decision be unrelated to the future path the economy would take under each alternative. *Overlap* requires every decision to have positive probability at every value of the covariates.

This thesis starts from the second assumption. Some policy actions are simply not on the table in some states of the economy. The Federal Reserve’s own documents record which: before every meeting, Board staff circulate the Bluebook, which lists the policy alternatives the Committee will consider. I use these *policy menus*, extracted from the Bluebooks by text analysis, to make three contributions.

1. **Structural non-overlap is observable.** In 133 scheduled meetings between August 1989 and January 2006 the FOMC never chose an action that was not on its Bluebook menu, and two thirds of the menus exclude either easing or tightening. The failure of overlap is not a feature of a fitted model. It is recorded in the Committee’s briefing documents.
2. **Menu-constrained propensity score and sharp bounds.** I estimate the decision model as an ordered probit renormalised over the menu, so that the propensity score is exactly zero for an action that is not on the menu. I then derive the sharp identified set for the average effect of a rate increase or decrease, exploiting the fact that in a menu that contains the status quo but not the action (or the reverse) half of the comparison is still identified. A one-parameter relaxation bounds how far effects in the non-overlap menus may differ from the effect in the overlap region, and yields a breakdown value for each estimate’s sign.
3. **Evidence.** Imposing the menus raises the log likelihood of the decision model by almost 19 points without any additional parameters. However, the effects that are point-identified on the overlap region imply that rate increases are followed by higher industrial production and lower unemployment, even when the propensity score conditions on the Greenbook forecasts that Romer and Romer (2004) use to purge policy of its predictable component. Worst-case bounds that make no assumption on the missing menus are wide and contain zero.

The negative result is informative. It shows that correcting the overlap problem, which is real and documented, does not deliver credible effects of monetary policy under selection on observables in this sample: what remains is a failure of unconfoundedness, most plausibly because the Committee acts on information about the future economy that neither macro aggregates, market expectations nor staff forecasts fully capture. The bounds framework makes this transparent, because it separates what the data identify from what must be assumed.

Related literature. The potential-outcomes approach to monetary policy follows Angrist and Kuersteiner (2011), Angrist et al. (2018) and Rambachan and Shephard (2019). The consequences of limited overlap for IPW are studied by Khan and Tamer (2010), Crump et al. (2009), Ma and Wang (2020) and, with many covariates, D’Amour et al. (2021); I study the case in which overlap fails exactly and the region of failure is observed. The bounds follow Manski (1990, 2003), and inference follows Imbens and Manski (2004). The text data build on the FOMC text-analysis project of Montiel Olea and coauthors and relate to Hansen et al. (2018). The use of staff forecasts to control for the Federal Reserve’s information set follows Romer and Romer (2004).

Outline. Section 2 describes the FOMC’s decision process and the construction of the policy menus. Section 3 sets out the framework and derives the bounds. Section 4 covers estimation and inference, and Section 5 reports results. Section 6 concludes.

2 Data: Policy Menus from the Bluebooks

2.1 How the FOMC prepares a decision

The Federal Open Market Committee (FOMC) meets eight times a year at scheduled meetings and occasionally by conference call. Each scheduled meeting is preceded by a structured set of staff documents (Dankner and Luecke, 2005). About two weeks before the meeting the *Beige Book* summarises anecdotal evidence from the twelve Reserve Bank districts. About a week before, the *Greenbook* gives the Board staff’s forecast for the domestic and international economy. On the Friday before the meeting the *Bluebook* sets out the policy alternatives for the Committee’s consideration, usually labelled A, B and C, each with a proposed federal funds rate target and a rationale. The meeting agenda then includes a discussion of these alternatives before the Committee votes. The format of these documents has changed over time: documents have been introduced, merged and retired since the corpus begins in 1936. Over the sample used here the sequence above is stable.

Two of these documents enter the analysis. The Bluebook defines the *menu* of actions the Committee considered, which is the object of this thesis, and the Greenbook provides the staff forecasts used as covariates in Section 4.

2.2 Extracting the menus

The Bluebooks are published with a five-year lag on the Board’s website as HTML and PDF files. Building on the FOMC text-analysis project of Montiel Olea, Giesecke, Chitale and Frermann, the documents were converted to raw text and every sentence mentioning a policy alternative was extracted with the regular expression

$$([\^{\cdot}]*)(\text{alternatives?}\backslash\text{s+}[a-e])(\backslash\cdot|[\^{\cdot}a-z]\backslash\cdot|[\^{\cdot}a-z][\^{\cdot}\backslash\cdot]+\backslash\cdot)$$

which captures a sentence containing “alternative” followed by a letter. The extracted sentences were then read and classified by hand, because the same target is sometimes described in different ways under different rationales. For each alternative I record the proposed change in the federal funds rate target, and classify it as easing (*E*), unchanged (*U*) or tightening (*T*).

The Committee’s actual decision is taken from the target changes in the replication files of Angrist et al. (2018). The sample runs from August 1989 to January 2006, the Greenspan chairmanship

after the Committee had begun to target the federal funds rate explicitly (Romer and Romer, 2004). It contains 133 scheduled meetings with a Bluebook menu. Target changes of 25, 50 and 75 basis points are pooled into the three categories because 50 and 75 basis point moves are too rare to estimate separately.

2.3 What the menus show

Table 1 cross-tabulates the menu offered with the decision taken. Two facts stand out. First, the decision is *always* on the menu: no meeting in the sample took an action that the Bluebook did not propose. The zeros in Table 1 are therefore structural, and they are observed rather than inferred. Second, most menus are restricted: 87 of 133 meetings (65%) were offered a menu without easing or without tightening, and only 46 had all three options. The full menu is also where the Committee was least likely to move. Only one of the 27 tightenings in the sample came from a full menu.

Table 1: Bluebook menus and FOMC decisions, August 1989 – January 2006

Bluebook menu X_t	Ease	Unchanged	Tighten	Meetings
$\{E\}$	3	0	0	3
$\{U\}$	0	1	0	1
$\{T\}$	0	0	7	7
$\{U, E\}$	17	16	0	33
$\{U, T\}$	0	24	19	43
$\{U, T, E\}$	9	36	1	46
Total	29	77	27	133

Notes: Scheduled FOMC meetings. Rows give the set of alternatives proposed in the Bluebook (E ease, U unchanged, T tighten); columns give the decision. Every decision lies in the menu.

Figure 1 shows where in the macroeconomic state space these restrictions bind. Panel (a) plots lagged inflation and industrial-production growth at each meeting, marking the meetings at which the Committee tightened or eased. Panel (b) marks instead the meetings whose menu *excluded* easing or tightening. Menus without easing are concentrated where output growth is above its sample average (88% of them), and menus without tightening where it is below (78%). The same pattern appears in the decisions themselves, but the menu shows it without relying on a fitted model: in about half of the meetings with below-average output growth (29 of 56) tightening was not even proposed, and the Committee tightened at only three of them. This is exactly the kind of covariate region in which a propensity score must be zero, and it is why a standard IPW estimator, which divides by the propensity score, is not available there.

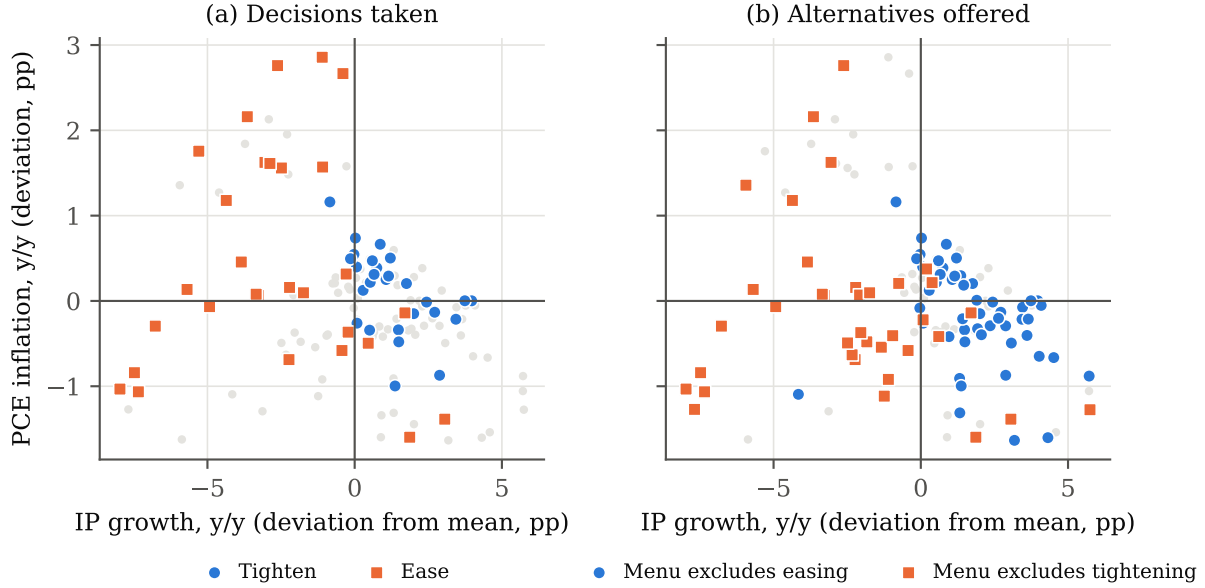


Figure 1: Policy decisions and policy menus in the inflation–output space. Each point is a scheduled FOMC meeting; axes show year-on-year PCE inflation and industrial-production growth in the previous month, as deviations from their 1989–2006 means. Grey points are meetings not in the highlighted group.

3 Framework: Policy Evaluation When Overlap Fails by Design

3.1 Treatment, menus and potential outcomes

Index scheduled FOMC meetings by t . The policy decision is $D_t \in \mathcal{D} = \{-1, 0, 1\}$ (ease, leave unchanged, tighten the federal funds rate target). The Bluebook menu is $X_t \in \mathcal{X}$, a non-empty subset of $\mathcal{D}$; in the data $\mathcal{X} = \{\{-1\}, \{0\}, \{1\}, \{-1, 0\}, \{0, 1\}, \{-1, 0, 1\}\}$. Other information available to the Committee (lagged macroeconomic data, market expectations, staff forecasts) is collected in $Z_t \in \mathbb{R}^k$, and $W_t = (X_t, Z_t)$.

For an outcome series y (log industrial production, the unemployment rate, the log PCE price index), let $Y_{t,h} = y_{t+h} - y_t$ be its cumulative change over the h months following the meeting. Each decision d induces a potential outcome $Y_{t,h}(d)$, and $Y_{t,h} = \sum_{d \in \mathcal{D}} \mathbf{1}\{D_t = d\} Y_{t,h}(d)$. The parameter of interest is the average dynamic effect of an increase ($d = 1$) or a decrease ($d = -1$) relative to leaving the target unchanged,

$$\theta_d(h) = \mathbb{E}[Y_{t,h}(d) - Y_{t,h}(0)], \quad d \in \{-1, 1\}, \quad h = 1, \dots, 24. \quad (1)$$

Expectations are taken over the stationary distribution of meetings. I drop the t and h subscripts when they are not needed. The propensity score is $p_d(w) = \mathbb{P}(D = d \mid W = w)$.

3.2 Assumptions

Assumption 1 (Selection on observables). $Y_h(d) \perp\!\!\!\perp D \mid W$ for all $d \in \mathcal{D}$ and all h .

Assumption 2 (Menu overlap). For all $w = (x, z)$ in the support of W : $p_d(x, z) > 0$ if and only if $d \in x$.

Assumption 3 (Bounded outcomes). $Y_h(d) \in [K_{1h}, K_{2h}]$ almost surely, for all d and h , with known $K_{1h} < K_{2h}$.

Assumption 1 is the identifying assumption of Angrist et al. (2018). It says that once we condition on what the Committee knew, including what it was offered, the remaining variation in its decision is as good as random with respect to the economy's future path. Assumption 2 replaces the usual overlap condition, $p_d(w) > 0$ for all d and w . It allows the propensity score to be zero, but only in the way the Bluebook dictates: the Committee does not take an action that is not on the menu (which holds without exception in the data, Table 1), and any action on the menu has some chance of being taken. The “only if” part is testable and holds. The “if” part is an assumption. Assumption 3 is a support restriction of the kind used by Manski (1990). It bites only on the cells in which the data are silent.

3.3 What the data identify

Write $\pi_x = \mathbb{P}(X = x)$ and $\mu_d(x) = \mathbb{E}[Y(d) \mid X = x]$. Conditioning on the menu,

$$\theta_d = \sum_{x \in \mathcal{X}} \pi_x (\mu_d(x) - \mu_0(x)). \quad (2)$$

Under Assumptions 1–2, for any $a \in x$ the conditional mean $\mu_a(x)$ is identified by inverse probability weighting within the menu:

$$\mu_a(x) = \mathbb{E} \left[\frac{\mathbf{1}\{D = a\} Y}{p_a(X, Z)} \mid X = x \right], \quad a \in x, \quad (3)$$

because $p_a(x, Z) > 0$ for every Z when $a \in x$. When $a \notin x$, the data never show $Y(a)$ in menu x and $\mu_a(x)$ is not identified. For a given d , partition the menus into four groups:

$$\begin{aligned} \mathcal{A}_d &= \{x : d \in x, 0 \in x\}, & \mathcal{B}_d &= \{x : 0 \in x, d \notin x\}, \\ \mathcal{C}_d &= \{x : d \in x, 0 \notin x\}, & \mathcal{N}_d &= \{x : d \notin x, 0 \notin x\}. \end{aligned}$$

For $d = 1$ these are $\mathcal{A}_1 = \{\{0, 1\}, \{-1, 0, 1\}\}$, $\mathcal{B}_1 = \{\{0\}, \{-1, 0\}\}$, $\mathcal{C}_1 = \{\{1\}\}$ and $\mathcal{N}_1 = \{\{-1\}\}$. The overlap region $\mathcal{A}_d$ is where the effect is point-identified. In $\mathcal{B}_d$ the status-quo mean is identified but the treated mean is not; in $\mathcal{C}_d$ the reverse; in $\mathcal{N}_d$ neither.

Proposition 1 (Sharp identified set). *Under Assumptions 1–3, the identified set for $\theta_d(h)$ is the interval $[L_d, U_d]$ with*

$$\begin{aligned} L_d &= \sum_{x \in \mathcal{A}_d} \pi_x (\mu_d(x) - \mu_0(x)) + \sum_{x \in \mathcal{B}_d} \pi_x (K_1 - \mu_0(x)) \\ &\quad + \sum_{x \in \mathcal{C}_d} \pi_x (\mu_d(x) - K_2) + \sum_{x \in \mathcal{N}_d} \pi_x (K_1 - K_2), \\ U_d &= \sum_{x \in \mathcal{A}_d} \pi_x (\mu_d(x) - \mu_0(x)) + \sum_{x \in \mathcal{B}_d} \pi_x (K_2 - \mu_0(x)) \\ &\quad + \sum_{x \in \mathcal{C}_d} \pi_x (\mu_d(x) - K_1) + \sum_{x \in \mathcal{N}_d} \pi_x (K_2 - K_1). \end{aligned}$$

Its width is $U_d - L_d = (K_2 - K_1) [\pi(\mathcal{B}_d) + \pi(\mathcal{C}_d) + 2\pi(\mathcal{N}_d)]$.

The proof is in Appendix A. The width is always below the width of the no-information bound, $2(K_2 - K_1)$, so the set is informative whenever some menu contains both d and 0 or one of them. It is also narrower than the naive bound that treats every cell outside $\mathcal{A}_d$ as uninformative, whose width is $2(K_2 - K_1)[1 - \pi(\mathcal{A}_d)]$, because the identified half of each one-sided cell is used. The 2020 version of this thesis used the naive width.

Remark 1 (Additional covariates). Conditioning on Z in (3) is needed for Assumption 1, but the bounds only require the menu-level means. If X happens to contain all the information in Z that predicts D , the propensity score does not depend on Z and (3) reduces to a difference in means within menu; nothing in Proposition 1 changes (Appendix A).

3.4 A sensitivity parameter for the missing menus

The worst case in Proposition 1 lets the average effect in menus outside the overlap region be as extreme as the outcome's support permits. A more useful statement bounds how different those effects can be from the identified effect in the overlap region,

$$\theta_d^A = \mathbb{E}[Y(d) - Y(0) \mid X \in \mathcal{A}_d] = \frac{\sum_{x \in \mathcal{A}_d} \pi_x(\mu_d(x) - \mu_0(x))}{\pi(\mathcal{A}_d)}.$$

Assumption 4 (λ -bounded heterogeneity). *For some $\lambda \geq 0$ and all $x \notin \mathcal{A}_d$: $|\mathbb{E}[Y(d) - Y(0) \mid X = x] - \theta_d^A| \leq \lambda$.*

Proposition 2. *Under Assumptions 1–4 the identified set is obtained from Proposition 1 by intersecting, for each $x \notin \mathcal{A}_d$, the cell-level interval with $[\theta_d^A - \lambda, \theta_d^A + \lambda]$. When the support constraint does not bind this is*

$$\Theta_d(\lambda) = \left[\theta_d^A - \lambda(1 - \pi(\mathcal{A}_d)), \theta_d^A + \lambda(1 - \pi(\mathcal{A}_d)) \right],$$

and the sign of θ_d is identified if and only if $\lambda < \lambda_d^ = |\theta_d^A| / (1 - \pi(\mathcal{A}_d))$.*

At $\lambda = 0$ effects are homogeneous across menus and $\theta_d = \theta_d^A$; as λ grows the set widens until it reaches the worst case. The *breakdown value* λ_d^* is the smallest degree of heterogeneity across menus that would overturn the sign of the estimated effect, measured in the units of the outcome. It gives a transparent way to judge how much the conclusion depends on extrapolating from the overlap region.

4 Estimation and Inference

4.1 A menu-constrained propensity score

Following Angrist and Kuersteiner (2011) and Angrist et al. (2018), the Committee's decision is modelled as an ordered probit in a latent policy index $Z'\beta$ with cutoffs $c_1 < c_2$, which gives unconstrained cell probabilities $\tilde{p}_{-1}(z) = \Phi(c_1 - z'\beta)$, $\tilde{p}_0(z) = \Phi(c_2 - z'\beta) - \Phi(c_1 - z'\beta)$ and $\tilde{p}_1(z) = 1 - \Phi(c_2 - z'\beta)$. To impose Assumption 2 exactly, the probabilities are renormalised over the menu:

$$p_d(x, z) = \frac{\tilde{p}_d(z) \mathbf{1}\{d \in x\}}{\sum_{d' \in x} \tilde{p}_{d'}(z)}. \quad (4)$$

This is the probability of choosing d from an ordered probit whose outcome is known to lie in x . It is zero exactly off the menu, it reduces to the ordered probit when $x = \mathcal{D}$, and it is degenerate at one in the single-option menus, which therefore do not enter the likelihood. The parameters (β, c_1, c_2) are estimated by maximum likelihood; standard errors come from the inverse Hessian, and average marginal effects use the delta method.

The 2020 version entered the menus as dummy variables in an unconstrained ordered probit, which keeps the probabilities off the menu small but positive; the renormalisation makes them exactly zero, as the bounds assume.

Covariates. I consider two information sets. The *macro* set follows Angrist et al. (2018): the surprise component of federal funds futures prices, two lags of monthly PCE inflation, two lags of the change in the unemployment rate, and two lags of year-on-year industrial-production growth. (The 2020 version used the contemporaneous level of the IP index; growth rates lagged to the meeting date are used here instead.) The *Greenbook* set, which is the main specification, replaces the macro variables with the staff forecasts the Committee had in hand at the meeting, as in Romer and Romer (2004): current- and next-quarter forecasts of real GDP growth and GDP-deflator inflation, and the current-quarter unemployment forecast, together with the futures surprise. Each meeting is matched to the most recent Greenbook released before it.

Table 2 reports average marginal effects on the probability of a rate increase. Columns (1)–(2) reproduce the progression of the 2020 thesis on the meeting sample: adding the futures surprise raises the log likelihood from -110.0 to -91.4 . Column (3) imposes the menus through (4), with the same covariates and no additional parameter, and the log likelihood rises by a similar amount again, to -72.6 . The text of the Bluebook carries as much information about the decision as market expectations do. Column (4), the main specification, uses the Greenbook information set and fits best (-52.7). Higher current-quarter forecasts of growth and inflation raise the probability of a hike. The next-quarter inflation forecast enters negatively once the current-quarter forecast is held fixed; the two are correlated (0.63), so their separate coefficients are less sharply determined than their sum. The futures surprise remains the strongest single predictor.

4.2 Estimating the bounds

Given fitted scores $\hat{p}$, the conditional means in (3) are estimated with normalised (Hájek) inverse probability weights within each menu,

$$\hat{\mu}_a(x) = \frac{\sum_{t: X_t=x} \mathbf{1}\{D_t = a\} Y_{t,h} / \hat{p}_a(W_t)}{\sum_{t: X_t=x} \mathbf{1}\{D_t = a\} / \hat{p}_a(W_t)},$$

and π_x by the sample share of meetings with menu x . Plugging these into Proposition 1 gives $\hat{L}_d$, $\hat{U}_d$ and $\hat{\theta}_d^A$. Normalised weights are bounded by the observed outcomes and are less sensitive to small scores than the unnormalised estimator. In the main specification the smallest estimated probability of an observed decision is 0.054, so no trimming is needed.

The outcome bounds K_{1h}, K_{2h} are set to the smallest and largest realised h -month change in each outcome over August 1989 – December 2010, a period that includes the 2001 and 2008–09 recessions. The assumption is that the potential outcomes of any single meeting stay within the range the economy actually went through over these two decades. The 2020 version calibrated K to the interquartile range of the estimated effects across horizons; calibrating it to the range of the outcome itself matches Assumption 3 and gives considerably wider bounds.

Table 2: Propensity-score models: average marginal effects on $\mathbb{P}(D_t = 1)$

	(1) Macro	(2) + FFF	(3) + Menus	(4) Greenbook
Inflation, lag 1	−0.056 (0.171)	−0.077 (0.150)	−0.013 (0.129)	
Inflation, lag 2	0.167 (0.151)	−0.039 (0.134)	−0.021 (0.114)	
Δ Unemployment, lag 1	−0.332 (0.203)	−0.239 (0.176)	−0.145 (0.142)	
Δ Unemployment, lag 2	−0.505** (0.226)	−0.239 (0.197)	−0.040 (0.159)	
IP growth, lag 1	0.075** (0.038)	0.059* (0.032)	0.041 (0.027)	
IP growth, lag 2	−0.037 (0.037)	−0.027 (0.032)	−0.022 (0.026)	
Fed funds futures surprise		0.738*** (0.119)	0.636*** (0.114)	0.575*** (0.082)
GB real GDP growth, q_0				0.049*** (0.014)
GB real GDP growth, q_1				0.019 (0.016)
GB inflation, q_0				0.057*** (0.022)
GB inflation, q_1				−0.097*** (0.026)
GB unemployment, q_0				0.005 (0.015)
Menu constraint	No	No	Yes	Yes
Log likelihood	−109.95	−91.41	−72.61	−52.65
Meetings	133	133	133	133

Notes: Ordered probit for the FOMC decision at 133 scheduled meetings, August 1989 – January 2006. Columns (3)–(4) renormalise the probabilities over the Bluebook menu, equation (4). Delta-method standard errors in parentheses. *, **, ***: significant at 10%, 5%, 1%. GB = Greenbook forecast; q_0 , q_1 = current and next quarter.

4.3 Inference

The estimators are smooth functions of sample moments and of the probit estimates, but the observations are serially dependent and the outcomes of nearby meetings overlap. I use a moving-block bootstrap over meetings (Künsch, 1989) with blocks of eight consecutive meetings (about one year) and $B = 499$ draws. Each draw re-estimates the propensity score, the menu shares and the conditional means, so first-stage uncertainty is propagated. The 2020 version left inference for future work because the conditional effects are computed on non-consecutive subsamples; resampling the full sequence of meetings and recomputing everything in each draw addresses that concern.

For θ_d^A I report the basic bootstrap 90% interval. For the partially identified θ_d I report the Imbens and Manski (2004) interval, which covers the true parameter (rather than the whole identified set) with probability 0.9 and remains valid when the identified set is narrow:

$$\left[\hat{L}_d - c_\alpha \hat{\sigma}_L, \hat{U}_d + c_\alpha \hat{\sigma}_U \right], \quad \Phi\left(c_\alpha + \frac{\hat{U}_d - \hat{L}_d}{\max(\hat{\sigma}_L, \hat{\sigma}_U)}\right) - \Phi(-c_\alpha) = 0.9,$$

with $\hat{\sigma}_L, \hat{\sigma}_U$ the bootstrap standard errors. The support bounds K are treated as known.

5 Results

Figure 2 and Table 3 report, for each outcome and each policy, three objects: the worst-case identified set of Proposition 1 (grey), the effect on the overlap region $\hat{\theta}_d^A$ with its 90% bootstrap interval (blue), and the breakdown value λ_d^* of Proposition 2. Responses are cumulative changes from the month of the meeting: percent for industrial production and the PCE price index, percentage points for the unemployment rate. The overlap region covers 67% of meetings for a rate increase (menus $\{U, T\}$ and $\{U, T, E\}$) and 59% for a decrease ($\{U, E\}$ and $\{U, T, E\}$). The main specification uses the Greenbook propensity score.

Worst-case bounds are not informative. With K equal to the range of outcomes the economy actually experienced, every identified set contains zero at every horizon, and so does every Imbens and Manski (2004) interval. Two years after a rate increase, the identified set for industrial production runs from -7.2% to $+4.3\%$. The menus that exclude the action or the status quo account for a third or more of all meetings, and without an assumption about them the data cannot sign the average effect. The 2020 version reported much narrower bounds because it calibrated K differently (Appendix C).

The overlap-region effects have the wrong sign for real activity. Where both the action and the status quo were on the menu, the effects are point-identified and estimated with some precision. After a rate *increase*, industrial production is higher than under no change (by 1.5% at six months and 1.9% at twelve, with 90% intervals excluding zero through 18 months) and unemployment is lower (by 0.6 percentage points at twelve months). After a rate *decrease*, industrial production is lower by 1.4% at twelve months and 2.2% at twenty-four, and unemployment is higher by 0.65 and 1.1 points. These are the opposite of the responses implied by standard theory and found by most of the literature, and they are the same in the macro specification (Appendix B).

Staff forecasts remove the price puzzle. The price responses behave differently. With the macro information set, a rate increase is followed by significantly *higher* prices from five months

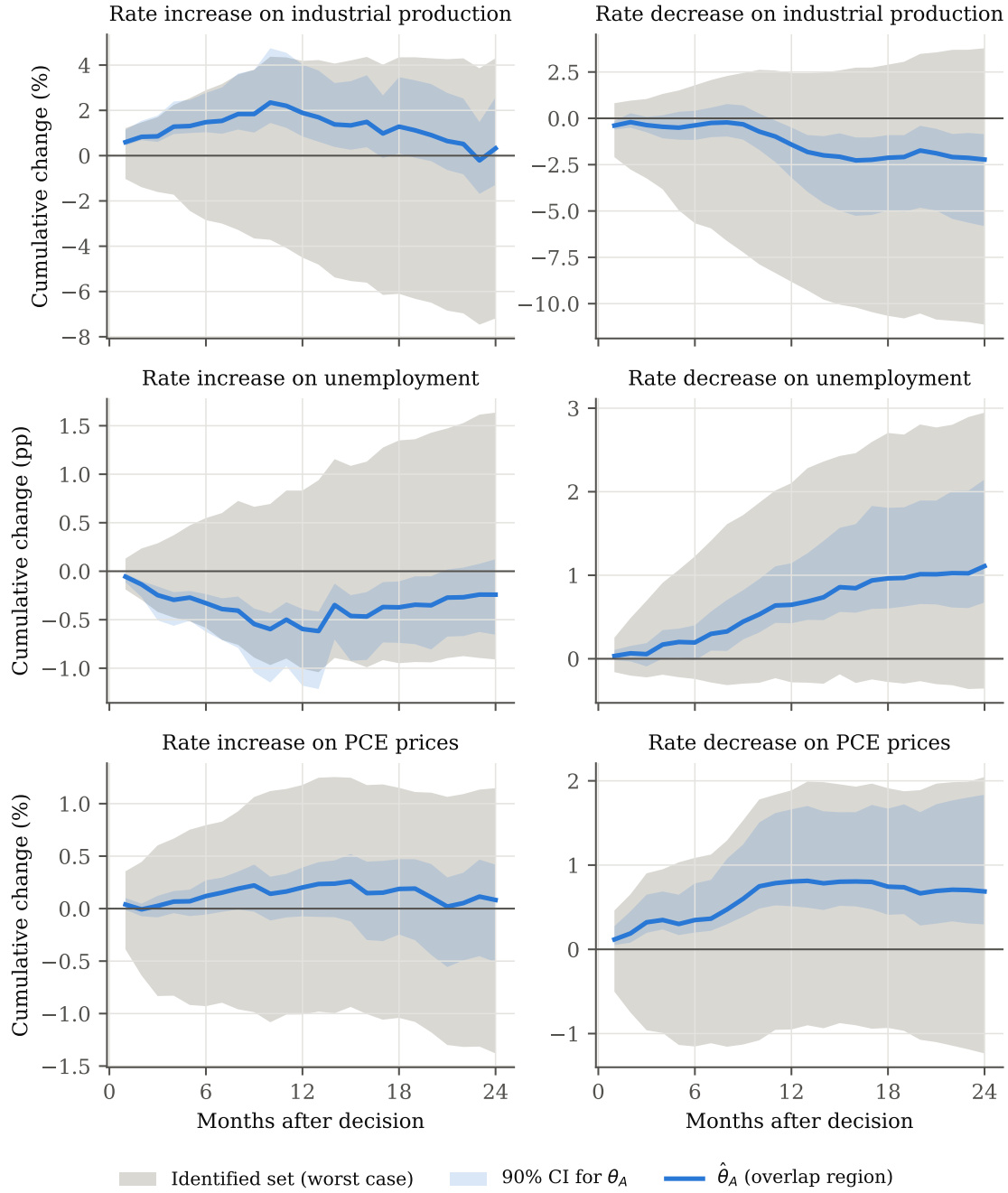


Figure 2: Cumulative effects of a rate increase (left) and decrease (right) relative to no change. Grey: worst-case identified set (Proposition 1). Blue: effect on the overlap region $\hat{\theta}_d^A$ with 90% moving-block bootstrap interval. Greenbook propensity score; 133 meetings, 1989–2006.

on, which is the well-known price puzzle (Sims, 1992). Conditioning on the Greenbook forecasts makes the price response to a rate increase small and statistically indistinguishable from zero at every horizon (0.20% at twelve months, interval $[-0.07, 0.39]$). This is the pattern Romer and Romer (2004) report: the puzzle reflects the Committee’s reaction to inflation it foresees, and the staff forecast captures that information. A rate decrease is followed by higher prices under both specifications (0.8–1.0% at twelve months), the direction theory predicts.

Table 3: Effects of rate changes relative to no change, Greenbook information set

Policy	h	$\hat{\theta}_A$	90% CI	Identified set	λ^*
<i>Industrial production (%)</i>					
Rate increase	6	1.48	[1.03, 2.77]	$[-2.8, 2.9]$	4.47
	12	1.89	[0.85, 4.02]	$[-4.5, 4.2]$	5.71
	18	1.28	[0.10, 3.47]	$[-6.1, 4.3]$	3.88
	24	0.32	$[-1.29, 2.58]$	$[-7.2, 4.3]$	0.97
Rate decrease	6	-0.38	$[-1.17, 0.41]$	$[-5.7, 1.8]$	0.93
	12	-1.42	$[-3.20, -0.49]$	$[-8.8, 2.5]$	3.49
	18	-2.12	$[-4.98, -0.90]$	$[-10.7, 2.9]$	5.23
	24	-2.22	$[-5.82, -0.85]$	$[-11.1, 3.8]$	5.48
<i>Unemployment rate (pp)</i>					
Rate increase	6	-0.33	$[-0.62, -0.24]$	$[-0.6, 0.5]$	1.00
	12	-0.60	$[-1.18, -0.39]$	$[-1.0, 0.8]$	1.80
	18	-0.37	$[-0.74, -0.11]$	$[-0.9, 1.3]$	1.12
	24	-0.24	$[-0.65, 0.12]$	$[-0.9, 1.6]$	0.73
Rate decrease	6	0.19	$[-0.02, 0.40]$	$[-0.2, 1.2]$	0.48
	12	0.65	$[0.42, 1.14]$	$[-0.3, 2.1]$	1.59
	18	0.96	$[0.60, 1.81]$	$[-0.3, 2.7]$	2.37
	24	1.11	$[0.67, 2.15]$	$[-0.4, 2.9]$	2.73
<i>PCE price index (%)</i>					
Rate increase	6	0.12	$[-0.06, 0.27]$	$[-0.9, 0.8]$	0.36
	12	0.20	$[-0.07, 0.39]$	$[-1.0, 1.2]$	0.61
	18	0.19	$[-0.25, 0.47]$	$[-1.0, 1.2]$	0.57
	24	0.08	$[-0.51, 0.42]$	$[-1.4, 1.1]$	0.25
Rate decrease	6	0.35	$[0.20, 0.78]$	$[-1.2, 1.1]$	0.86
	12	0.80	$[0.51, 1.66]$	$[-1.0, 1.9]$	1.98
	18	0.74	$[0.41, 1.67]$	$[-0.9, 1.9]$	1.83
	24	0.69	$[0.30, 1.84]$	$[-1.2, 2.0]$	1.69

Notes: Cumulative change h months after the meeting. $\hat{\theta}_A$: effect on meetings whose menu contains both the action and no change, with basic moving-block bootstrap 90% interval (499 draws, blocks of 8 meetings). Identified set: worst-case bounds on the average effect over all meetings (Proposition 1). λ^* : smallest heterogeneity between the non-overlap menus and the overlap region, in the outcome’s units, that would make the sign of the average effect unidentified (Proposition 2).

How much extrapolation is needed? If effects were the same in every menu ($\lambda = 0$), the overlap-region estimates would be the average effects. The breakdown values in Table 3 say how far that homogeneity can be relaxed before the sign of the average effect is lost. For a rate increase, a third of the meetings are outside the overlap region, so $\lambda^* = 3|\hat{\theta}^A|$: the twelve-month IP response keeps its sign unless the average effect in the menus without the status quo or without tightening differs from the overlap-region effect by more than 5.7 percentage points. For a rate decrease two fifths of meetings lie outside the region and $\lambda^* \approx 2.4|\hat{\theta}^A|$. Figure 3 traces the full set as a function of λ . Reconciling the average effects with the textbook signs would require the effect in the

non-overlap menus to be not merely different from, but several times larger than and opposite in sign to, the effect in the overlap region. That cannot be ruled out, but it would be an odd defence: the counter-theoretical signs come from the meetings where the data are most informative, not from extrapolation.

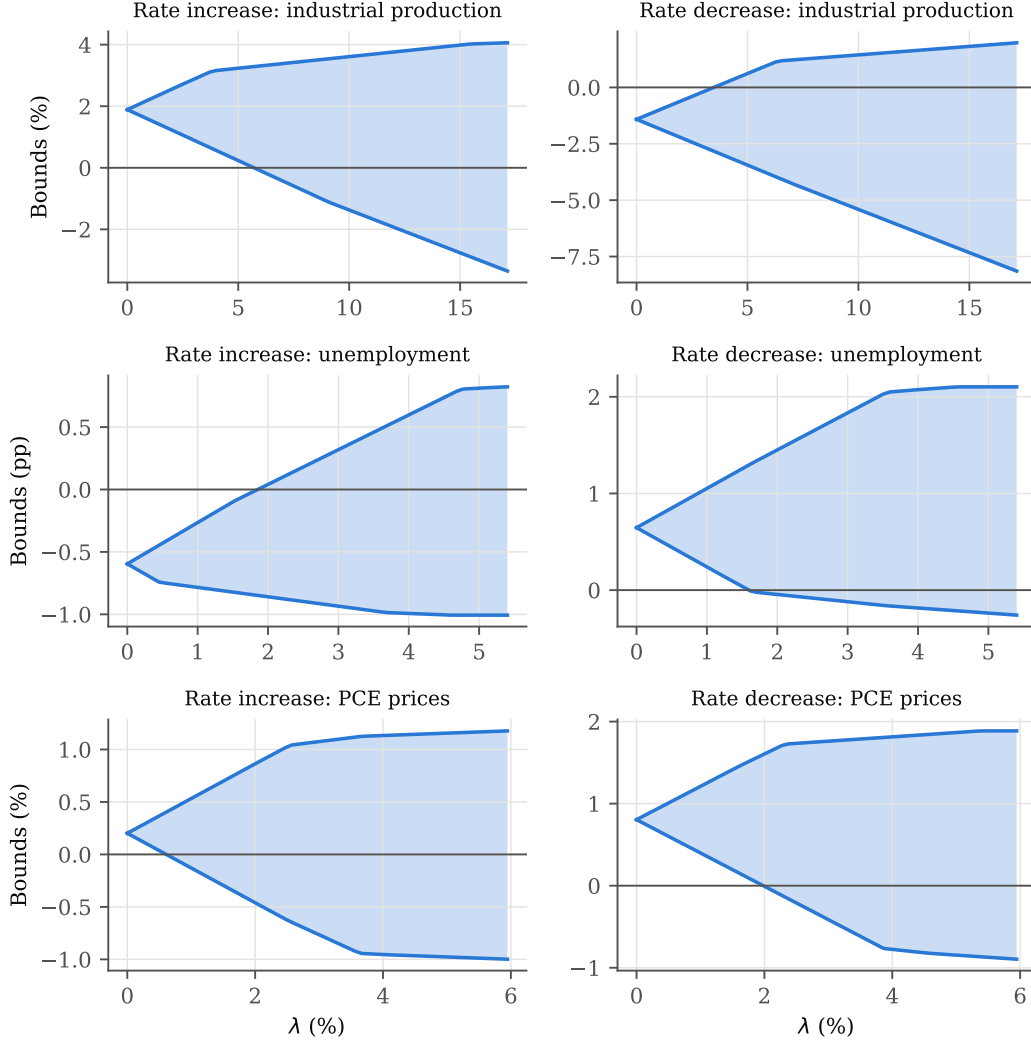


Figure 3: Identified set for the twelve-month effect as a function of λ , the maximal difference between the average effect in a non-overlap menu and in the overlap region (Assumption 4). At $\lambda = 0$ the set is the point $\hat{\theta}_d^A$; for large λ it reaches the worst case. Greenbook propensity score.

Interpretation. The overlap-region estimates are credible only under Assumption 1. Their signs indicate that it fails. The Committee tightens when it expects the economy to strengthen, and the expected strength shows up in subsequent output whether or not the Committee acts. Neither lagged data, market expectations nor the Greenbook forecasts absorb this completely. Two features of the sample make the problem hard to escape. First, the identifying variation is thin: all but one of the 27 tightenings come from $\{U, T\}$ menus, so the effect of a rate increase is essentially a comparison of hikes and holds within those 43 meetings. Second, weighting within a menu only reweights

meetings the Committee itself judged similar enough to offer the same options, so the remaining differences between hikes and holds are exactly the judgemental information that is hardest to measure. Asymmetries between increases and decreases, which the 2020 version emphasised following [Hamilton and Jordà \(2002\)](#) and [Angrist and Kuersteiner \(2011\)](#), cannot be separated from this selection problem with the present data.

6 Conclusion

This thesis set out to relax one of the two assumptions behind semiparametric estimates of the effects of monetary policy. Its central observation holds up: the Federal Reserve’s own briefing documents show that some actions are not on the table at some meetings, and the Committee never chose an action its staff had not proposed. Treating the Bluebook menu as a covariate that dictates where the propensity score is zero gives a decision model that fits markedly better at no cost in parameters, and a sharp characterisation of what the data can and cannot say about the average effect of policy.

What the data can say is limited. Without restrictions on the menus that exclude the action or the status quo, the average effect is not signed. With effects assumed similar across menus, the estimates are precise but have the wrong sign for output and unemployment, even with the Greenbook forecasts in the propensity score. The forecasts do remove the price puzzle for rate increases. The contribution of the menu is therefore to isolate the problem. Overlap can be handled; unconfoundedness is what fails.

Three extensions follow naturally. First, the menus can be extended to the Bernanke and Yellen years and to the size of the proposed moves, increasing the identifying variation. Second, the Bluebook’s own description of each alternative, and the transcripts, could supply covariates for the Committee’s judgemental information, the component that current controls miss. Third, the bounds framework can be combined with assumptions weaker than selection on observables, such as monotone treatment response in the spirit of [Manski \(2003\)](#), which would turn the menu restrictions into informative bounds without requiring unconfoundedness.

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A Proofs

Proof of Proposition 1. By the law of iterated expectations, $\theta_d = \sum_x \pi_x(\mu_d(x) - \mu_0(x))$, and π_x is identified.

Identification of $\mu_a(x)$ for $a \in x$. By Assumption 2, $p_a(x, z) > 0$ for all z in the support of $Z \mid X = x$. By Assumption 1, $\mathbb{E}[\mathbf{1}\{D = a\}Y \mid X = x, Z] = \mathbb{E}[\mathbf{1}\{D = a\}Y(a) \mid X = x, Z] = p_a(x, Z) \mathbb{E}[Y(a) \mid X = x, Z]$. Dividing by $p_a(x, Z)$ and integrating over $Z \mid X = x$ gives (3).

Non-identification of $\mu_a(x)$ for $a \notin x$. By Assumption 2, $\mathbb{P}(D = a \mid X = x) = 0$, so the distribution of observables (Y, D, X, Z) places no restriction on the conditional distribution of $Y(a)$ given $X = x$ beyond Assumption 3. In particular, for any $m \in [K_1, K_2]$ the joint distribution obtained by setting $Y(a) = m$ almost surely on $\{X = x\}$ generates the same observables, satisfies

Assumption 1 (a constant is independent of D), and has $\mu_a(x) = m$. Hence the identified set for $\mu_a(x)$ is exactly $[K_1, K_2]$.

Sharpness of the sum. The unidentified means $\{\mu_a(x) : a \notin x\}$ enter θ_d linearly with coefficients $\pm\pi_x$, and each can be set independently to either endpoint by the construction above, applied menu by menu. The minimum of θ_d sets $\mu_d(x) = K_1$ for $x \in \mathcal{B}_d \cup \mathcal{N}_d$ and $\mu_0(x) = K_2$ for $x \in \mathcal{C}_d \cup \mathcal{N}_d$; the maximum does the reverse. This gives L_d and U_d . Every intermediate value is attained by convex combination, so the identified set is the interval $[L_d, U_d]$. Subtracting yields the width. $\square$

Proof of Proposition 2. Assumption 4 adds, for each $x \notin \mathcal{A}_d$, the constraint $\mu_d(x) - \mu_0(x) \in [\theta_d^A - \lambda, \theta_d^A + \lambda]$, where θ_d^A is identified. Intersecting with the cell-level intervals of Proposition 1 and summing gives the set. If no support constraint binds, each cell contributes $\pi_x(\theta_d^A \pm \lambda)$, so the bounds are $\pi(\mathcal{A}_d)\theta_d^A + (1 - \pi(\mathcal{A}_d))(\theta_d^A \pm \lambda) = \theta_d^A \pm \lambda(1 - \pi(\mathcal{A}_d))$, which excludes zero iff $\lambda < |\theta_d^A|/(1 - \pi(\mathcal{A}_d))$. $\square$

Information in X versus Z . If the menu contains all the information in Z that predicts D , i.e. $p_a(x, z) = p_a(x)$, then (3) becomes $\mu_a(x) = \mathbb{E}[Y \mid D = a, X = x]$, a difference in means within menu, and conditioning on Z is harmless. In general $\mathbb{E}[Y(a) \mid X = x] = \int \mathbb{E}[Y(a) \mid X = x, Z = z] dF_{Z|X=x}(z)$: the integral is over the distribution of Z *conditional on the menu*. The 2020 version integrated over the marginal distribution of Z , which is incorrect when X and Z are dependent, as they are here.

B Robustness: macro information set

Table 4 and Figure 4 repeat the main results with the macro information set of Angrist et al. (2018) (column (3) of Table 2) in place of the Greenbook forecasts.

Table 4: Effects of rate changes, macro information set

Policy	h	$\hat{\theta}_A$	90% CI	Identified set	λ^*
<i>Industrial production (%)</i>					
Rate increase	6	1.36	[0.78, 2.67]	[−2.9, 2.8]	4.11
	12	1.87	[0.62, 4.21]	[−4.5, 4.2]	5.67
	18	1.11	[−0.39, 3.54]	[−6.2, 4.3]	3.35
	24	0.38	[−1.41, 3.28]	[−7.1, 4.4]	1.14
Rate decrease	6	−0.41	[−1.40, 0.49]	[−5.7, 1.7]	1.02
	12	−1.68	[−3.85, −0.67]	[−9.1, 2.2]	4.14
	18	−2.33	[−5.60, −0.86]	[−11.0, 2.5]	5.73
	24	−2.52	[−6.62, −0.76]	[−11.7, 3.3]	6.21
<i>Unemployment rate (pp)</i>					
Rate increase	6	−0.35	[−0.65, −0.21]	[−0.6, 0.5]	1.05
	12	−0.73	[−1.41, −0.49]	[−1.1, 0.7]	2.21
	18	−0.52	[−1.01, −0.24]	[−1.1, 1.2]	1.56
	24	−0.57	[−1.30, −0.27]	[−1.1, 1.4]	1.74
Rate decrease	6	0.20	[−0.09, 0.42]	[−0.2, 1.2]	0.49
	12	0.73	[0.52, 1.34]	[−0.3, 2.1]	1.80
	18	1.13	[0.78, 2.19]	[−0.2, 2.8]	2.78
	24	1.27	[0.83, 2.56]	[−0.3, 3.0]	3.13
<i>PCE price index (%)</i>					
Rate increase	6	0.24	[0.08, 0.44]	[−0.9, 0.9]	0.71
	12	0.36	[0.08, 0.62]	[−0.9, 1.3]	1.09
	18	0.62	[0.21, 1.11]	[−0.7, 1.4]	1.86
	24	0.72	[0.20, 1.37]	[−0.9, 1.6]	2.17
Rate decrease	6	0.46	[0.23, 0.93]	[−1.0, 1.2]	1.13
	12	0.95	[0.52, 1.83]	[−0.8, 2.0]	2.34
	18	0.99	[0.49, 2.02]	[−0.7, 2.2]	2.44
	24	1.04	[0.45, 2.33]	[−0.9, 2.4]	2.55

Notes: As Table 3, with the propensity score of Table 2, column (3).

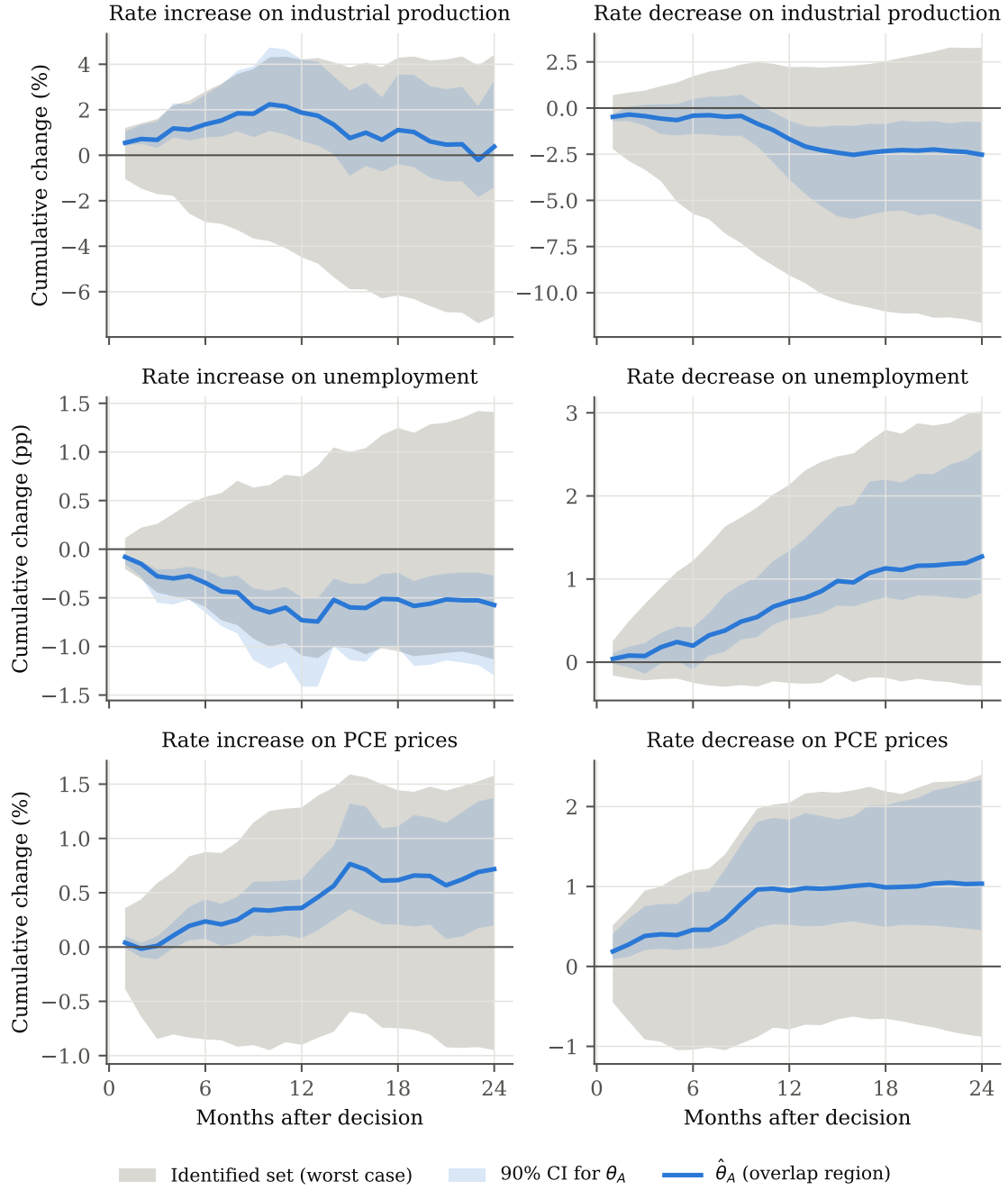


Figure 4: Cumulative effects of rate changes, macro information set. See Figure 2.

C Comparison with the 2020 version

The revision changes the estimation in several ways, described in the main text: the propensity score now imposes the menu’s zeros and uses covariates observed at the time of the meeting (Section 4); the bounds use the identified half of one-sided menus (Section 3); the outcome bounds K are calibrated to the range of the outcome rather than to the spread of the estimates; cumulative effects are bounded directly; and inference is reported. The 2020 estimation also applied a scaling step to the point-identified component of some bounds that is not part of the estimator derived in Section 3; the revision drops it. Most of the difference in the real-activity results between the two versions comes from this step.